

```
pipeline {
    agent any

    environment {
        DOCKER_IMAGE = "itsmekalai/conference-app:latest"
    }

    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url: 'https://github.com/itsmekalai15/kalai.git'
            }
        }

        stage('Docker Build') {
            steps {
                script {
                    echo "Building Conference Webpage Image..."
                    bat "docker build -t ${DOCKER_IMAGE} ."
                }
            }
        }

        stage('Docker Push') {
            steps {
                script {
                    withCredentials([usernamePassword(credentialsId: 'dockerhub-creds',
usernameVariable: 'DOCKER_USER', passwordVariable: 'DOCKER_PASS')]) {
```

```

        bat 'docker login -u %DOCKER_USER% -p %DOCKER_PASS%'

        bat "docker push ${DOCKER_IMAGE}"

    }

}

}

}

stage('K8s Deployment') {
    steps {
        script {
            echo "Deploying Conference App to Kubernetes..."

            // Loads the secret file with ID 'itsmekalai' into the KUBE_CFG variable
            withCredentials([file(credentialsId: 'itsmekalai', variable: 'KUBE_CFG')]) {
                // Passes the securely loaded file to the kubectl command
                bat 'kubectl apply -f deployment.yaml --kubeconfig="%KUBE_CFG%"'
            }
        }
    }
}

post {
    success {
        echo "CI/CD Pipeline Complete. Access via NodePort 30007"
    }
}
}

```

Deployment.yaml

apiVersion: apps/v1

kind: Deployment

metadata:

name: conference-webapp

labels:

app: conference

spec:

replicas: 2

selector:

matchLabels:

app: conference

template:

metadata:

labels:

app: conference

spec:

containers:

- name: web-container

image: itsmekalai/conference-app:latest # Replace with your DockerHub tag

ports:

- containerPort: 80

apiVersion: v1

kind: Service

metadata:

name: conference-service

spec:

type: NodePort

selector:

app: conference

ports:

- port: 80

targetPort: 80

nodePort: 30007 # The application will be visible at <NodeIP>:30007

maven docker (change folder name-evenodd)

Dockerfile:

FROM eclipse-temurin:21

WORKDIR /app

COPY target/evenodd-app-1.0-SNAPSHOT.jar app.jar

CMD ["java", "-jar", "app.jar"]

Pom.xml :(maven)

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">
```

```
  <modelVersion>4.0.0</modelVersion>
```

```
  <groupId>com.example</groupId>
```

```
  <artifactId>javainheritance</artifactId>
```

```
  <packaging>jar</packaging>
```

```
  <version>1.0-SNAPSHOT</version>
```

```
  <name>javainheritance</name>
```

```
<properties>
```

```
<maven.compiler.source>1.8</maven.compiler.source>  
<maven.compiler.target>1.8</maven.compiler.target>  
</properties>
```

```
<dependencies>  
  <!-- Updated to JUnit 5 to match your "org.junit.jupiter" imports -->  
  <dependency>  
    <groupId>org.junit.jupiter</groupId>  
    <artifactId>junit-jupiter-api</artifactId>  
    <version>5.10.0</version>  
    <scope>test</scope>  
  </dependency>  
  <dependency>  
    <groupId>org.junit.jupiter</groupId>  
    <artifactId>junit-jupiter-engine</artifactId>  
    <version>5.10.0</version>  
    <scope>test</scope>  
  </dependency>  
</dependencies>
```

```
<build>  
  <plugins>  
    <plugin>  
      <groupId>org.apache.maven.plugins</groupId>  
      <artifactId>maven-surefire-plugin</artifactId>  
      <version>3.1.2</version>  
    </plugin>  
  </plugins>
```

</build>

</project>

Maven docker: (Change folder name)

FROM eclipse-temurin:21

WORKDIR /app

COPY target/javaproject-app-1.0-SNAPSHOT.jar app.jar

CMD ["java", "-jar", "app.jar"]

Static docker:

FROM nginx:alpine

COPY . /usr/share/nginx/html

EXPOSE 80

Step 2 - Create Simple Java Application (in local machine)

The folder structure will be, (To be created in the same order)

simple-java-app

```
|
|
|— src
|   |
|   |— main
|   |   |
|   |   |— java
|   |       |
|   |       |— App.java
|   |
|   |
```

```
|   |— test
|       |
|       |— java
|           |
|           |— AppTest.java
|
|— pom.xml
|— Dockerfile
|— deployment.yaml
```